

OOPS ASSIGNMENT – 07

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1.

```
#include <iostream>
```

```
using namespace std;
```

```
class Polygon
```

```
{
```

```
public:
```

```
    int width, height;
```

```
    void set_value(int w, int h)
```

```
{
```

```
    width = w;
```

```
    height = h;
```

```
}
```

```
    virtual int area()
```

```
{
```

```
    return 0;
```

```
}
```

```
};
```

```
class Rectangle : public Polygon
```

```
{
```

```
public:
    int area()
    {
        return width * height;
    }
};
```

```
class Triangle : public Polygon
{
public:
    int area()
    {
        return width * height / 2;
    }
};
```

```
int main()
{
    Polygon *p;
    Rectangle r;
    Triangle t;

    p = &r;
    p->set_value(4,5);
    cout << p->area() << endl;

    p = &t;
```

```

p->set_value(4,5);

cout << p->area() << endl;


return 0;

}

```

```

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20
10
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```

2.

```

#include using namespace std;

class Shape { public: virtual void area() { }

virtual void display()
{
}

};

class Circle : public Shape { public: int r;

Circle(int x)
{
    r = x;
}

void area()
{
    cout << 3.14 * r * r << endl;
}

void display()
{
    cout << "Circle" << endl;
}

```

```
};
```

```
class Rectangle : public Shape { public: int l, b;
```

```
Rectangle(int x, int y)
```

```
{  
    l = x;  
    b = y;  
}
```

```
void area()
```

```
{  
    cout << l * b << endl;  
}
```

```
void display()
```

```
{  
    cout << "Rectangle" << endl;  
}
```

```
};
```

```
class Triangle : public Shape { public: int h, b;
```

```
Triangle(int x, int y)
```

```
{  
    h = x;  
    b = y;  
}
```

```
void area()
```

```
{  
    cout << 0.5 * h * b << endl;  
}
```

```
void display()
```

```
{  
    cout << "Triangle" << endl;  
}
```

```

};

int main() { Shape *s;

Circle c(5);
Rectangle r(4,6);
Triangle t(3,4);

s = &c;
s->display();
s->area();

s = &r;
s->display();
s->area();

s = &t;
s->display();
s->area();

return 0;

}

```



```

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Circle
78.5
Rectangle
24
Triangle
6
(base) vihaan@Vihaans-Macbook 10.GRAPH5 %

```

3.

```

#include using namespace std;

class Triangle { public: double area(int b, int h) { return 0.5 * b * h; }

double area(int a)
{
    return (1.732/4) * a * a;
}

double area(int a, int b, int h)

```

```
{  
    return 0.5 * b * h;  
}
```

```
};
```

```
int main() { Triangle t;
```

```
    cout << t.area(4,5) << endl;
```

```
    cout << t.area(6) << endl;
```

```
    cout << t.area(5,5,4) << endl;
```

```
    return 0;
```

```
}
```

```
• (base) vihaan@Vihaans-Macbook 10.GRAPH5 % cd "/Users/vihaan/Desktop/DSA/10.GRAPH5/" && g++ trash.cpp -o trash && "/Users/vihaan/Desktop/DSA/10.GRAPH5/"trash  
10  
15.588  
10  
○ (base) vihaan@Vihaans-Macbook 10.GRAPH5 %
```

4.

```
#include <iostream>
```

```
using namespace std;
```

```
class Student
```

```
{
```

```
public:
```

```
    virtual void show() = 0;
```

```
};
```

```
class Engineering : public Student
```

```
{
```

```
public:
```

```
void show()
{
    cout << "Engineering" << endl;
}
};
```

```
class Medicine : public Student
{
public:
    void show()
    {
        cout << "Medicine" << endl;
    }
};
```

```
class Science : public Student
{
public:
    void show()
    {
        cout << "Science" << endl;
    }
};
```

```
int main()
{
    Student *s[3];
```

Engineering e;

Medicine m;

Science sc;

s[0] = &e;

s[1] = &m;

s[2] = ≻

for (int i = 0; i < 3; i++)

{

s[i]->show();

}

return 0;

}

```
● (base) vihaan@Vihaans-Macbook 10.GRAPHs % cd "/Users/vihaan/Desktop/DSA/10.GRAPHs/" && g++ trash.cpp -o trash && "/Users/vihaan/Desktop/DSA/10.GRAPHs/"trash
Engineering
Medicine
Science
○ (base) vihaan@Vihaans-Macbook 10.GRAPHs %
```

5.

#include <iostream>

using namespace std;

class Time

{

private:

int h, m, s;

public:

Time()

{

h = m = s = 0;

}

Time(int hh, int mm, int ss)

{

h = hh;

m = mm;

s = ss;

}

Time operator+(Time t)

{

Time temp;

temp.s = s + t.s;

temp.m = m + t.m + temp.s / 60;

temp.s = temp.s % 60;

temp.h = h + t.h + temp.m / 60;

temp.m = temp.m % 60;

return temp;

}

void show()

{

```

        cout << h << ":" << m << ":" << s << endl;
    }
};

```

```

int main()
{
    Time t1(5,15,34), t2(9,53,58), t3;
    t3 = t1 + t2;
    t3.show();

    return 0;
}

```

```

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15:9:32
○ (base) vihaan@Vihaans-Macbook 10.GRAPH5 %

```

6.

```

#include <iostream>
using namespace std;

```

```

class STRING
{
public:
    string s;

    STRING(string str)
    {
        s = str;
    }
}

```

```
bool operator==(STRING t)
```

```
{  
    return s == t.s;  
}
```

```
STRING operator+(STRING t)
```

```
{  
    return STRING(s + t.s);  
}
```

```
void show()
```

```
{  
    cout << s << endl;  
}  
};
```

```
int main()
```

```
{  
    STRING s1("Hello"), s2("World"), s3("");
```

```
    if (s1 == s2)
```

```
        cout << "Equal" << endl;
```

```
    else
```

```
        cout << "Not Equal" << endl;
```

```
    s3 = s1 + s2;
```

```
    s3.show();
```

```
return 0;
```

```
}
```

```
● (base) vihaan@Vihaans-Macbook 10. GRAPHS % cd "/Users/vihaan/Desktop/DSA/10. GRAPHS/" && g++ trash.cpp -o trash && "/Users/vihaan/Desktop/DSA/10. GRAPHS/"trash
Not Equal
HelloWorld
○ (base) vihaan@Vihaans-Macbook 10. GRAPHS %
```

7.

```
#include <iostream>
```

```
using namespace std;
```

```
class Matrix
```

```
{
```

```
public:
```

```
    int a[2][2];
```

```
void input()
```

```
{
```

```
    for(int i=0;i<2;i++)
```

```
        for(int j=0;j<2;j++)
```

```
            cin >> a[i][j];
```

```
}
```

```
void show()
```

```
{
```

```
    for(int i=0;i<2;i++)
```

```
    {
```

```
        for(int j=0;j<2;j++)
```

```
            cout << a[i][j] << " ";
```

```
        cout << endl;
```

```
    }  
}
```

```
friend Matrix operator*(Matrix m1, Matrix m2);  
};
```

```
Matrix operator*(Matrix m1, Matrix m2)
```

```
{  
    Matrix res;  
  
    for(int i=0;i<2;i++)  
    {  
        for(int j=0;j<2;j++)  
        {  
            res.a[i][j] = 0;  
            for(int k=0;k<2;k++)  
            {  
                res.a[i][j] += m1.a[i][k] * m2.a[k][j];  
            }  
        }  
    }  
}
```

```
    return res;  
}
```

```
int main()
```

```
{
```

```
Matrix m1, m2, m3;
```

```
m1.input();
```

```
m2.input();
```

```
m3 = m1 * m2;
```

```
m3.show();
```

```
return 0;
```

```
}
```

```
● cd "/Users/vihaan/Desktop/DSA/10.GRAPHs/" && g++ trash.cpp -o trash && "/Users/vihaan/Desktop/DSA/10.GRAPHs/"trash
264269112 399152896
267075489 -1212861456
○ (base) vihaan@Vihaans-Macbook 10.GRAPHs %
```

8.

```
#include using namespace std;
```

```
class Array { public: int a[5];
```

```
int& operator[](int i)
```

```
{
```

```
    if (i < 0 || i >= 5)
```

```
    {
```

```
        cout << "Out of bounds" << endl;
```

```
        exit(0);
```

```
    }
```

```
    return a[i];
```

```
}
```

```
};
```

```
int main() { Array arr;
```

```

for(int i=0;i<5;i++)
    arr[i] = i + 1;

for(int i=0;i<5;i++)
    cout << arr[i] << " ";

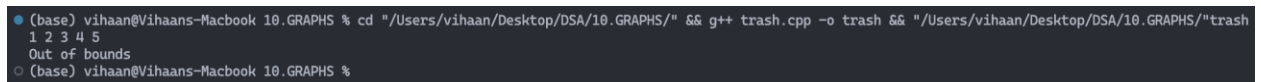
cout << endl;

cout << arr[10];

return 0;

}

```



```

(base) vihaan@Vihaans-Macbook 10.GRAPHs % cd "/Users/vihaan/Desktop/DSA/10.GRAPHs/" && g++ trash.cpp -o trash && "/Users/vihaan/Desktop/DSA/10.GRAPHs/"trash
1 2 3 4 5
Out of bounds
(base) vihaan@Vihaans-Macbook 10.GRAPHs %

```

9.

```

#include <iostream>

using namespace std;

```

```

class Test
{
public:

    void operator()(int n)
    {
        cout << "One value: " << n << endl;
    }

    void operator()(int a, int b)
    {
        cout << "Two values: " << a << " " << b << endl;
    }
}

```

```

    }

    void operator()(int a, int b, int c)
    {
        cout << "Three values: " << a << " " << b << " " << c << endl;
    }
};

```

```

int main()
{
    Test t;

    t(5);
    t(1,2);
    t(3,4,5);

    return 0;
}

```

```

● (base) vihaan@Vihaans-Macbook 10.GRAPHs % cd "/Users/vihaan/Desktop/DSA/10.GRAPHs/" && g++ trash.cpp -o trash && "/Users/vihaan/Desktop/DSA/10.GRAPHs/"trash
One value: 5
Two values: 1 2
Three values: 3 4 5
○ (base) vihaan@Vihaans-Macbook 10.GRAPHs %

```

10.

```

#include <iostream>

using namespace std;

```

```

class Test
{
public:

```

```

int x;

friend istream& operator>>(istream &in, Test &t)
{
    in >> t.x;
    return in;
}

friend ostream& operator<<(ostream &out, Test &t)
{
    out << t.x;
    return out;
}
};

int main()
{
    Test t;

    cin >> t;
    cout << t;

    return 0;
}

```

```

● cd "/Users/vihaan/Desktop/DSA/10.GRAPHs/" && g++ trash.cpp -o trash && "/Users/vihaan/Desktop/DSA/10.GRAPHs/"trash
0%
○ (base) vihaan@Vihaans-Macbook 10.GRAPHs %

```

```
#include <iostream>

using namespace std;

class Test
{
private:
    float x;

public:
    Test(float f)
    {
        x = f;
    }

    void show()
    {
        cout << x << endl;
    }
};
```

```
int main()
{
    float a = 5.5;

    Test t = a;

    t.show();
}
```

```
return 0;
```

```
}
```

```
• (base) vihaan@Vihaans-Macbook 10.GRAPH5 % cd "/Users/vihaan/Desktop/DSA/10.GRAPH5/" && g++ trash.cpp -o trash && "/Users/vihaan/Desktop/DSA/10.GRAPH5/"trash 5.5
○ (base) vihaan@Vihaans-Macbook 10.GRAPH5 %
```

12.

```
#include <iostream>
```

```
using namespace std;
```

```
class Test
```

```
{
```

```
public:
```

```
    float x;
```

```
    Test(float a)
```

```
    {
```

```
        x = a;
```

```
    }
```

```
    operator float()
```

```
    {
```

```
        return x;
```

```
    }
```

```
};
```

```
int main()
```

```
{
```

```
    Test t(5.5);
```

```
float a;
```

```
a = t;
```

```
cout << a << endl;
```

```
return 0;
```

```
}
```

```
● (base) vihaan@Vihaans-Macbook 10. GRAPHS % cd "/Users/vihaan/Desktop/DSA/10. GRAPHS/" && g++ trash.cpp -o trash && "/Users/vihaan/Desktop/DSA/10. GRAPHS/"trash
5.5
○ (base) vihaan@Vihaans-Macbook 10. GRAPHS %
```

13.

```
#include <iostream>
```

```
#include <math.h>
```

```
using namespace std;
```

```
class Polar
```

```
{
```

```
public:
```

```
float r, theta;
```

```
Polar(float a, float b)
```

```
{
```

```
    r = a;
```

```
    theta = b;
```

```
}
```

```
};
```

```
class Cartesian
```

```

{
public:
    float x, y;

    Cartesian(Polar p)
    {
        x = p.r * cos(p.theta);
        y = p.r * sin(p.theta);
    }

    void show()
    {
        cout << x << " " << y << endl;
    }
};

int main()
{
    Polar p(10,5);
    Cartesian c = p;
    c.show();

    return 0;
}

```

```

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2.83662 -9.58924
○ (base) vihaan@Vihaans-Macbook 10. GRAPHS %

```